

SmartLearn Institute is a private educational organization planning to develop and launch an Online Learning Management System (LMS) to support its students and faculty. The system will allow students to access course materials, submit assignments, attend live classes, and view results online. Faculty members will be able to upload lectures, manage assignments, mark attendance, and generate student performance reports.

The management has allocated a 6-month project duration and a dedicated project team consisting of: Project Manager, System Analyst, UI/UX Designer, Software Developers, Database Administrator, Testing Team

Before starting development, the project manager decides to apply Work Breakdown Structure (WBS) to break the project into manageable components for better planning, scheduling, cost estimation, and task allocation.

Project Scope Details

The LMS project will involve the following major activities:

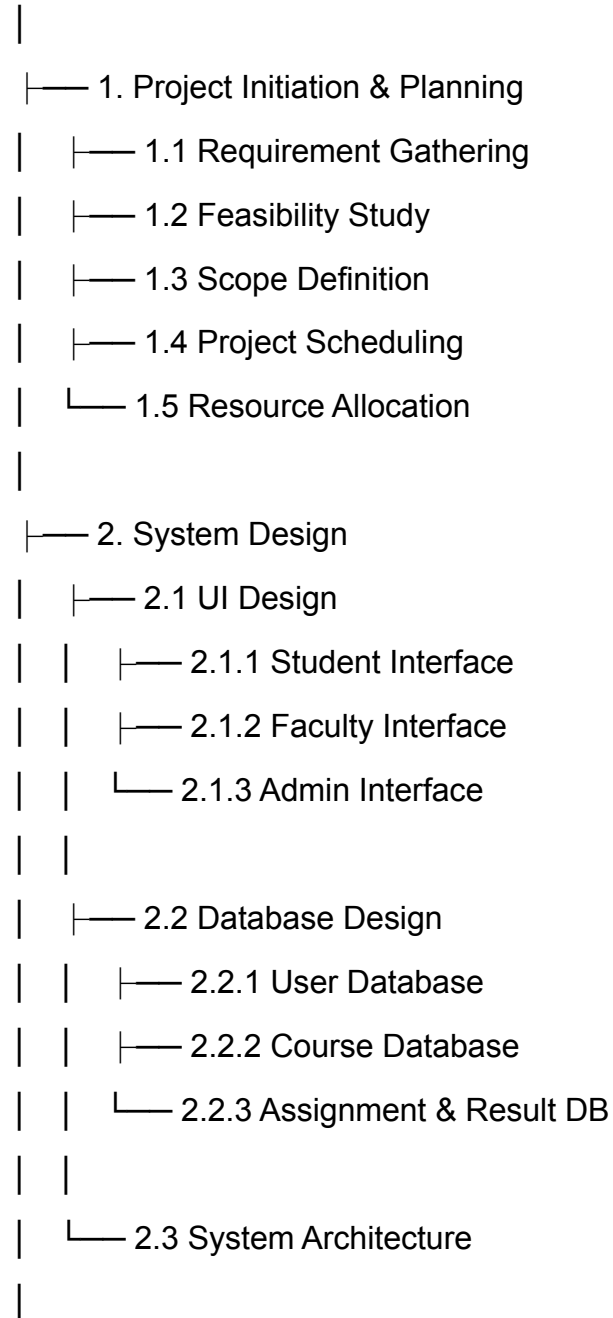
- Project initiation and planning, including requirement analysis, feasibility study, project scheduling, and team allocation.
- System design, including user interface design, database design, and overall system architecture.
- System development, which includes:
 - Student module (registration, login, course access, assignment submission)
 - Faculty module (content upload, assignment creation, attendance management)
 - Admin module (user management, course management, report generation)
- Testing activities including unit testing, integration testing, system testing, and user acceptance testing.
- Deployment and implementation, including system installation, data migration, and user training.
- Maintenance and support, including bug fixing, system updates, and performance monitoring after launch.

The project manager wants to ensure that all deliverables are clearly identified, responsibilities are assigned properly, and no task is overlooked during execution.

Question

- a) Using the above case study, **develop a detailed Work Breakdown Structure (WBS)** up to at least **three levels**.
- b) Clearly show **major deliverables and sub-tasks** involved in the LMS project.

Online Learning Management System (LMS)



|—— 3. System Development

| |—— 3.1 Student Module

| | |—— 3.1.1 Registration & Login

| | |—— 3.1.2 Course Enrollment

| | |—— 3.1.3 Access Materials

| | |—— 3.1.4 Assignment Submission

| | |—— 3.1.5 View Results

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| |—— 3.2 Faculty Module

| | |—— 3.2.1 Faculty Login

| | |—— 3.2.2 Upload Content

| | |—— 3.2.3 Create Assignments

| | |—— 3.2.4 Attendance Management

| | |—— 3.2.5 Grading

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| |—— 3.3 Admin Module

| |—— 3.3.1 User Management

| |—— 3.3.2 Course Management

| |—— 3.3.3 System Monitoring

| |—— 3.3.4 Report Generation

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|—— 4. Testing

| |—— 4.1 Unit Testing

| |—— 4.2 Integration Testing

| |—— 4.3 System Testing

| |—— 4.4 User Acceptance Testing

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|—— 5. Deployment & Implementation

| |—— 5.1 System Installation

| |—— 5.2 Data Migration

| |—— 5.3 User Training

| |—— 5.4 Go-Live

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└—— 6. Maintenance & Support

|—— 6.1 Bug Fixing

|—— 6.2 System Updates

|—— 6.3 Performance Monitoring

└—— 6.4 Technical Support